

Sarthak Consul

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Education

Indian Institute of Technology Bombay, Mumbai, India
Bachelor of Technology (with Honours) in Electrical Engineering

July 2016 - Present

- **Major GPA:** 9.62/10
- **Minor** in Computer Science and Engineering (Minor GPA: 9.5/10)

Research Internships & Projects

Segmentation of Lacunar Objects from Ultra High Resolution μ CT Bone Scans

ETH Zürich

Guide: *Prof. Dr. Ralph Müller, D-HEST, Institute for Biomechanics*

Summer 2019

- Worked on extracting lacunar structures from trabecular bone samples of rare disease patients
- Modified Fully Convolutional Networks like WNet for segmentation in 3D images in an unsupervised setting
- Worked on a 2 stage method that extracts regions using adaptive thresholding followed by a classifier to fine-tune results by removing noise structures, achieving an **F1 Score** of **98.4%**

Semantic Segmentation of Medical Images

IIT Bombay

Guide: *Prof. Amit Sethi, EE, Medical Deep learning and AI Lab*

Summer 2018

- Tested ability of Fully Connected Convolutional Networks in segmentation tasks in medical images.
- Implemented Global Convolution on a fully-connected ResNet backbone using Pytorch for the **segmentation of lungs** from chest x-rays and **nuclei segmentation** from h/e stained histopathological sections of organs
- Part of a five member team testing various architectures and conventional image processing techniques as post-processing to improve the state of the art results of **nuclei segmentation** to a mean F1 score of **85%**
- Reviewed research papers on **ReNet** and **ReSeg**, **MobileNet**, and **Siamese Neural Networks**
- **Mentored** two freshmen in the team on the basics of machine learning and deep learning
- Work was used to host the MoNuSeg challenge at MICCAI 2018.
- Annotated and pre-processed data from around 800 patients affected by oral cancer

Scholastic Achievements

- Awarded **Advanced Performer** grade (AP) in **4 courses**: *Biology* (15 out of 445), *Introduction to Electrical Systems* (2 out of 149), *Network Theory* (10 out of 143), and *Electronic Devices Lab* (3 out of 139)
- Granted a **Change of Branch** based on excellent academic standing in freshmen year [2016]
- Secured All India Rank 840 (out of around 200,000 candidates) in JEE Advanced [2016]
- Recipient of KVPY Fellowship (**AIR 16**) by Dept. of Science and Technology, Govt. of India [2014]
- Placed in **National Top 1%** in National Standard Examination in Junior Sciences [2012]
- Ranked **4th** in State Science Talent Search Examination (SSTSE), conducted by Govt. of Rajasthan [2014]
- Recipient of **NTSE** Scholarship by National Council of Research and Training [2012]

Miscellaneous Projects

Digitally Programmable Analog Computer

EE344: Electronics Design Lab

Prof. *Mukul Chandorkar*, EE, IIT Bombay

Spring 2019

Designed and developed a standalone digitally programmable analog computer, capable of solving non-linear dynamical systems upto the fifth order, demonstrating hardware-in-loop capabilities. The system is capable of working in real-time, fully programmable and features on-board power management [Technical Report]

Human Detection from Post-disaster LWIR Imagery

EE338: Digital Signal Processing

Prof. *Vikram M. Gadre*, EE, IIT Bombay

Spring 2019

- Designed and implemented a system to detect humans from low-wavelength infrared (LWIR) images for the purposes of streamlining rescue operations
- Used a two-stage approach of extracting regions using **MSER**, followed by using **Integral Channel Features** such as local sums, **Histogram of Gradients** along with the **DCT** as features to classify using a **SVM**.
- Presented to students and teachers of various tier 2 and 3 colleges across the country as part of TEQIP-KIT-2019 workshop under the initiative of the Ministry of Human Resource Development (MHRD), Govt. of India

Audio Source Separation using a Low Latency Neural Network

CS419: Machine Learning

Prof. *Sunita Sarawagi*, CSE, IIT Bombay

Fall 2018

- Developed a novel low-latency neural network to separate the different musical instruments from music
- Used depthwise separable convolutions, inspired by the **GCN** architecture to reduce parameters.

Tree based Stereo Matching for Postcapture Refocus

Prof. Suyash Awate & Prof. Ajit Rajwade, CSE, IIT Bombay

CS663: Digital Image Processing

Fall 2018

- Implemented a Hybrid tree based stereo matching algorithm which calculates the depth of images using Minimum spanning trees.
- Incorporated CLMF (Cross Local Multipoint filter) to use as a edge preserving smoothener for refining the depth map used the depth estimates to interactively refocus and generate depth blurring.

Multicycled and Pipelined Implementation of IITB-RISC ISA

Prof. Virendra Singh, EE, IIT Bombay

EE309: Microprocessors

Fall 2018

- Designed and implemented a 6-stage pipelined multicycle RISC processor in VHDL, consisting of arithmetic, logical and branching instructions, and tested on Altera DE0-Nano FPGA board.
- Implemented Branch Predictors, Priority Encoders and Hazard Detection Units to reduce latencies and significantly reduce the branch hazards.

Analog Sound Locator

Prof. Siddharth Tallur, EE, IIT Bombay

EE230: Analog Systems Lab

Spring 2018

- Designed and implemented an analog circuit to locate distant sound sources. The setup mimicked bi-aural mammalian ears, using the phase difference of the received sound signal to locate the source.
- Amplitude invariance** was achieved using JFET delay circuits with dynamic phase locking
- Circuit was compact and had a fast convergence (spatial angle was obtained **within a millisecond**)
- One of the **winning projects** in the final evaluation (*amongst 68 teams*)

Reaction Game

Prof. Madhav P. Desai, EE, IIT Bombay

EE224: Digital Systems Lab

Spring 2018

- Built a game to measure reaction times to visual stimulus using a Krypton (Altera MAX V) CPLD board
- Modelled the game as a **finite-state machine** and realised it with **register-transfer level** abstraction in VHDL
- Incorporated false-react detection, timeouts and debouncing features. Used the **Double Dabble Algorithm** to convert the binary reaction times to decimal. Integrated a LCD controller to give real-time scores on a LCD

Pulse Rate Sensor Design

Prof. Siddharth Tallur, EE, IIT Bombay

EE236: Electrical Devices Lab

Autumn 2017

- Created a pulse-rate monitor circuit - variation of blood flow is observed by change in reflectivity of a finger
- Detection was done with the help of a PT333 phototransistor (paired with an IR LED). The input signal is sent through a Op-Amp based filtering scheme (band pass filter with a comparator) to minimize noise.

Building a Cryptosuite

Prof. Bernard L. Menezes, CSE, IIT Bombay

CS 101: Computer Programming

Autumn 2016

- Designed a program that encrypted and decrypted alphanumeric messages using **RSA Algorithm**
- Computed Discrete logarithms using **Baby Step Giant Step Algorithm**, binary and prime field operations, and modulo arithmetic (*addition, multiplication, exponentiation by binary exponentiation, and inverse*)

Mentorship Experience

Teaching Assistant

Prof. Swapneel Mahajan, MA, IIT Bombay

MA207: Partial Differential Equations

Fall 2018

- Taught and held doubt-clearing sessions for a class of approximately 60 students. Involved in the formulation of examinations, grading of answer scripts and other logistics.

Instructor of Machine Learning Bootcamp

Summer 2018

- Taught the fundamentals of Machine Learning to a class of approximately **130** students
- Covered the basics of ML before delving into the theory and implementation of popular algorithms of supervised and unsupervised learning such as **Random Forest Algorithm**, **CNNs**, and **k-means Clustering**.
- Prepared Jupyter notebooks (in Python), meant to serve as tutorials, for the various algorithms discussed

Department Academic Mentor

2019-20

Mentoring a second year student with a view to help him cope academically

Technical Skills

Programming Languages - C, C++, MATLAB, Python, Bash, SQL, VHDL, Assembly and Embedded C

Software libraries - PyTorch, TensorFlow, Keras

Web Development - HTML5, CSS3, PHP

Simulation Tools - NgSpice, Simulink, LabVIEW, Wireshark, ns-3

Design Software - EAGLE, AutoCAD, SolidWorks, Photoshop, L^AT_EX

Hardware - Arduino, Intel 8081, Intel 8085, Altera DE0-Nano

Relevant Coursework

Electrical Engineering

Microprocessors, EM Waves, Probability and Random Processes, Network Theory, Microelectronics, Analog and Digital Systems, Digital Signal Processing, Digital and Analog Communication, Control Systems, Wavelets**

Mathematics Courses

Matrix Computations, Linear Algebra, Data Analysis, Calculus, Differential Equations, Complex Analysis

Computer Science and Engineering

Advanced Machine Learning*, Reinforcement Learning*, Automatic Speech Recognition*, Computer Vision, Machine Learning, Digital Image Processing, Medical Image Processing**, Advanced Image Processing**, Computer Networks, Data Structures and Algorithms, Operating Systems, Logic for CS*

Online Courses

- **Reinforcement Learning** (DeepMind)
- **Convolutional Neural Network for Computer Vision** (Stanford University - CS231n)
- **Machine Learning** (Stanford University - CS229 and Coursera)
- **Advanced MATLAB for Scientific Computing** (Stanford University - Lagunita) [with Distinction]
- **Neural Networks for Machine Learning** (University of Toronto - Coursera)

* Indicates courses to be completed by Dec 2019

** Indicates courses to be completed by May 2020